visualization-drafts

```r
#Loading general use and cleaning packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(here)
```

```
here() starts at /Users/ej/GitHub/ncos-vegetation
```

```r
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
#Loading color palette package
library(NatParksPalettes)
```

```r
#Reading in vegetation survey data
veg <- read_csv(here("data", "veg.csv"))
```

```
Rows: 27036 Columns: 17
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (10): monitors, transect_name, vp_axis, site, transect_side, vp_zone, q...
dbl   (6): x1, transect_length, num_quad, transect_distance, percent_cover, ...
date  (1): date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#Reading in vegetation metadata
veg_metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

```
Rows: 257 Columns: 4
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): year_pool, transect_name
dbl (2): year, num_quad

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#Creating a dataset of dominant plant species in salt marsh habitat
salt_marsh_dominant_species <- veg |>
  #Cleaning column names
  clean_names() |>
  #Filtering to focus only on salt marsh habitat
  filter(habitat == "Salt Marsh") |>
  #Keeping only plant cover categories, not bare ground, thatch, or other cover
  filter(cover_category %in% c("NATIVE COVER", "NON-NATIVE COVER")) |>
  #Grouping by plant species and cover category
  group_by(psoc, cover_category) |>
  #Calculating total and mean percent cover for each species
  summarize(total_percent_cover = sum(percent_cover, na.rm = TRUE),
            mean_percent_cover = mean(percent_cover, na.rm = TRUE),
            n_observations = n()) |>
  #Ungrouping the dataframe
  ungroup() |>
```

```r
  #Ordering species from highest to lowest total cover
  arrange(desc(total_percent_cover)) |>
  #Keeping the top 10 dominant species
  slice_head(n = 10) |>
  #Cleaning species names for the figure
  mutate(species = str_replace_all(psoc, "_", " "),
         species = str_to_sentence(species),
         species = fct_reorder(species, total_percent_cover))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by psoc and cover_category.
i Output is grouped by psoc.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(psoc, cover_category))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```r
#Displaying the summarized dominant species data
salt_marsh_dominant_species
```

```
# A tibble: 10 x 6
   psoc     cover_category total_percent_cover mean_percent_cover n_observations
   <chr>    <chr>                        <dbl>              <dbl>          <int>
 1 Salicor~ NATIVE COVER                 31927              44.8             712
 2 Distich~ NATIVE COVER                  3942              21.2             186
 3 Parapho~ NON-NATIVE CO~                3444              12.7             271
 4 Plantag~ NON-NATIVE CO~                2913              18.6             157
 5 Jaumea_~ NATIVE COVER                  2490              15.9             157
 6 Polypog~ NON-NATIVE CO~                2334              11.6             202
 7 Franken~ NATIVE COVER                  2284               9.14            250
 8 Distich~ NATIVE COVER                  2274              22.7             100
 9 Arthroc~ NATIVE COVER                  1114              19.9              56
10 Spergul~ NATIVE COVER                   950               4.28            222
# i 1 more variable: species <fct>
```